

Assignment 1 Cryptography

Perform **encryption and decryption operations using AES** with different modes of operation using **OpenSSL** and **GPG** tools **openssl**

Use **OpenSSL** to generate random keys and initialization vectors (IV).

Using a sample plaintext file (for example plain.txt), perform **encryption and decryption** using all modes(ECB,CBC,CFB,OFB,CTR). [write a repeated sequence of data in plaintext like BBBB....] For each mode you must:

1. Encrypt the plaintext file.
2. Decrypt the encrypted file.
3. Verify that the decrypted file matches the original plaintext.

Encryption Using GPG

Use the **GPG tool** to perform symmetric encryption.

Tasks:

1. Encrypt a file using GPG symmetric encryption.
2. Specify the encryption algorithm (AES256)
3. Decrypt

Submit a report that includes:

- All commands used for encryption and decryption.
- Screenshots.
- Explanation of the output produced by each mode and tool.
- Observations about the differences between the AES modes and the behavior of OpenSSL and GPG.

Before you start

Change the hostname of Kali to your id

Try this command

```
(kali@kali)-[~]  
$ hostname  
kali  
  
(kali@kali)-[~]  
$ sudo hostnamectl set-hostname yourID  
[sudo] password for kali:
```

Replace **yourID** with your actual ID.

After running the command, open a **new terminal window** and you will notice that the hostname has changed.

```
File Actions Edit View Help  
(kali@yourID)-[~]  
$
```

All screenshots in your report must show your ID as the hostname.

If the hostname is not changed to your student ID, the submission will not be considered and you will receive zero.

The form

<https://forms.gle/46b1998T4KsKGMQZ6>

Deadline 19th March 2026